

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Own the end-to-end architecture of a multi-agent AI system built with LangGraph, the OpenAI Agents SDK, CrewAI, and AutoGen with MCP tool-calling, defining agent orchestration patterns for autonomous, multi-step workflows at 99%+ uptime across millions of daily transactions.
- Designed the distributed data and state architecture behind agent memory and context management via Redis checkpointing, so interrupted agentic workflows resume correctly from the last committed step at scale.
- Drove cross-team technical strategy for a deployment-gated evaluation pipeline governing agent and AI-generated output quality, raising the engineering bar by blocking 15% of failed releases before production.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Led the redesign of the RAG retrieval and data-preparation pipeline, improving recall 61% to 89% on a 2K-query eval set through hybrid dense-sparse retrieval and re-ranking over structured and unstructured data.
- Architected a model observability platform tracking drift, latency, and evaluation regressions across production agent/model outputs, establishing governance-aligned rollback workflows as an early engineer at this AI startup.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Led the architecture and design of 25+ production C++ and SQL-backed microservices in distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation and query logic, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning across the data layer.
- Led standardization of failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, mentoring engineers through code reviews and design write-ups, reducing production escalations 60% in one quarter.

Projects

LangFetch | tinyurl.com/LangFetch

- LangGraph agent compiling natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking; deployed as an MCP-compatible server for enterprise data queries.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, SQL, Java, TypeScript, C++

Agentic AI & LLM: Agentic AI, Multi-Agent Architectures, LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, MCP, RAG, Agent Memory & State Management

Architecture & Distributed Systems: System Design, Distributed Systems, Microservices, Data Modeling, REST APIs, Reliability & Scaling

Cloud & Data: AWS (Lambda, EC2, RDS, SQS, SageMaker), PostgreSQL, Vector Databases (Pinecone, FAISS), Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer